THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The point $(-6, -2)$ lies: 1

(a) Quadrant II **(b)** Quadrant I **(c)** Quadrant III **(d)** Quadrant IV

2. The distance between $A(1, 2)$ and $B(4, 6)$ is: 1

(a) 5 units **(b)** 7 units **(c)** 6 units **(d)** 4 units

3. The distance between $A(-4, -3)$ and $B(4, 3)$ is: 1

(a) 10 units **(b)** 12 units **(c)** 11 units **(d)** 9 units

4. The midpoint of the segment joining $A(-3, 5)$ and $B(7, -1)$ is: 1

(a) $(4, 4)$ **(b)** $(2, 2)$ **(c)** $(2, -2)$ **(d)** $(5, -3)$

5. $M(3, 2)$ is the midpoint of AB and $A(1, -4)$. The coordinates of B are: 1

(a) $(5, -8)$ **(b)** $(4, -2)$ **(c)** $(2, 6)$ **(d)** $(5, 8)$

6. The degree of the polynomial -9 is: 1

(a) 0 **(b)** 2 **(c)** 1 **(d)** 3

7. The value of the linear polynomial $2x + 3$ at $x = -6$ is: 1

(a) 9 **(b)** -7 **(c)** -15 **(d)** -9

8. The value of the linear polynomial $-2x - 1$ at $x = 5$ is: 1

(a) -11 **(b)** 11 **(c)** -9 **(d)** -13

9. The value of $2x^2 - 1x + 1$ at $x = -1$ is: 1

(a) 6 **(b)** 2 **(c)** 4 **(d)** 1

10. The sequence $1, 7, 13, 19, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $6n - 7$ **(b)** $n + 6$ **(c)** $6n + 1$ **(d)** $6n - 5$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the coordinates of the vertices of a rectangle with one vertex at the origin, length 7 units along the x-axis and breadth 4 units along the y-axis (in Quadrant I). 2

12. In which quadrant (or on which axis) do the points $(-2, 5)$, $(6, -3)$, $(0, -4)$ and $(-1, -1)$ lie? 2

13. A chess club charges ₹200 joining fee plus ₹50 per match. Write the expression for the amount paid for m matches and find m if a player paid ₹750. 2

14. Find the equation of the line through $(0,\ 3)$ with slope 2. Where does it cut the y-axis? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that $R(-5,\ -1)$, $B(-2,\ -5)$ and $C(4,\ -12)$ are not collinear. 3

16. The balance of a prepaid plan is ₹600 and it reduces by ₹15 daily. Write $b(x)$ and find after how many days it runs out. 3

17. Triangle ADM has vertices $A(3, 4)$, $D(7, 1)$, $M(9, 6)$. It is reflected in the y-axis. Write the image vertices and show that $A'D' = AD$. 3

OR

Find a point on the y-axis equidistant from $A(6,\ 5)$ and $B(-4,\ 3)$. 3

18. Without plotting, state the y-intercept and say which of the lines $y = 2x - 1$, $y = 2x + 1$ and $y = 2x + 5$ are parallel. Give a reason. 3

OR

Draw (describe the points for) the graphs of $y = \frac{1}{2}x$, $y = x$ and $y = 2x$ using suitable points. What does the value of a in $y = ax$ tell about the line? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that the points $A(2,\ -2)$, $B(-2,\ 1)$ and $C(5,\ 2)$ are the vertices of a right-angled triangle. Find its area. 5

OR

Case study: A drone flies from the point $A(1, 2)$ to $B(7, 10)$ and then to $C(7, 2)$ (units in 100 m). 5

(i) Find the distance flown from A to B.

(ii) Find the total distance $A \to B \to C$.

(iii) How much shorter is the direct path $A \to C$?

20. **Case study:** A farmer cuts a 300-ft fence into two pieces; the longer is four times the shorter. A rectangular plot has length 3 more than twice its width and perimeter 24 m. 5

(i) Find the two pieces.

(ii) Find the dimensions of the plot.

(iii) Find the area of the plot.

OR

Draw the graphs of $y = 3x$ and $y = -3x$ using suitable points. 5

(i) Where do they meet?

(ii) Compare their slopes.

(iii) State what the sign of the slope tells about growth and decay.

END OF QUESTION PAPER — Total: 40 Marks